Network Design

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1 Introduction

Modern agriculture increasingly relies on connected devices to improve monitoring and efficiency. The Internet of Things enables physical devices such as sensors and cameras to share data across networks and support farm management (Elijah et al., 2018). As connectivity grows, the network must be designed to remain stable, secure, and easy to manage across areas such as offices, retail points, greenhouses, and security systems.

This report presents the design and simulation of a farm network using Packet Tracer. Virtual Local Area Networks are used to create logical segmentation, with inter vlan routing enabling controlled communication. Both wired and wireless connectivity support user devices and IoT equipment across the farm.

Part A

2 Investigation

2.1 What is IoT and IoE

The Internet of Things refers to a network of physical devices connected to the internet that collect and exchange data without direct human involvement. In agriculture, IoT is widely used to monitor soil moisture, temperature, and irrigation, helping farm operators manage resources more efficiently (Elijah et al., 2018).
The Internet of Everything extends this idea by integrating people, data, processes, and devices into one system. While IoT focuses mainly on device communication, IoE enables more intelligent services by combining device data with human interaction and automated workflows (Chauhan and Jain, 2019).
In smart farming, IoT provides monitoring, and while IoE helps convert this data into effective management decisions.

2.2 Wireless Networks

Wireless networks are important in smart farming because they allow sensors and monitoring systems to communicate without physical cabling, supporting real time data collection and remote control.
Common IoT wireless technologies include WiFi, ZigBee, LoRaWAN, and 5G, which differ in range, power consumption, and bandwidth. WiFi provides high data rates but needs more infrastructure, while LoRaWAN offers longer range with lower power but reduced throughput (Rajasekaran and Anandamurugan, 2019).
In this project, WiFi was selected due to its wide availability, simple deployment, and sufficient bandwidth for office and monitoring needs, making it a practical choice for the simulated farm.

2.3 Network Organisation and Performance

Effective network organisation is essential when many devices share the same infrastructure. Without proper structure, networks may suffer from congestion and excessive broadcast traffic. Virtual Local Area Networks address this by logically dividing the physical network into smaller segments, which reduces unnecessary traffic and improves bandwidth utilisation (Shaik, 2024).
vlan also simplify management by grouping devices according to their function, making it easier to apply policies, monitor traffic, and isolate faults. In smart farm environments, this is particularly useful because areas such as offices, shop systems, greenhouse sensors, and security cameras generate different traffic patterns. Separating high traffic systems like cameras and “POS” devices from low bandwidth IoT sensors helps maintain stable network performance.

2.4 Vulnerability and Security

IoT environments are exposed to security risks because many devices have limited processing capability and weak built in protection. As a result, they can become entry points for unauthorised access or service disruption in agricultural systems. Virtual Local Area Networks improve network security by logically separating devices into different broadcast domains. This limits unnecessary traffic and helps contain potential attacks within each segment (Makeri et al., 2021). In this design, every functional area of the farm is placed in a dedicated VLAN. Greenhouse IoT sensors are isolated from office systems, and security cameras operate in their own segment, which reduces the overall attack surface while still allowing controlled inter vlans communication when required.

2.5 Commercial and Environmental Implications

IoT technologies offer clear commercial benefits in agriculture. Real time monitoring allows farmers to track soil conditions, temperature, and irrigation more accurately, reducing waste and improving resource efficiency. Automated systems such as smart irrigation can apply inputs only when needed, lowering costs and potentially increasing crop yield (Elijah et al., 2018).
From an environmental perspective, precision monitoring improves the use of water, fertilisers, and energy, which helps reduce the environmental impact of farming and supports more sustainable practices (Agrimonti et al., 2021).

However, IoT adoption also presents challenges. Initial investment costs can be high for small farms, and concerns around device security and data privacy must be managed carefully. Despite these issues, IoT based smart farming remains a practical and valuable approach when properly implemented (Elijah et al., 2018).

Part B

3 Network Design

3.1 Design Overview

The proposed network supports the daily operations of a smart farm including the office, shop, PYO kiosk, greenhouses, and security systems. Because each area has different traffic and security needs, the network was designed to remain organised, reliable, and scalable.

The design follows a hierarchical model with one central router, one core switch, and four distribution switches. Inter vlan routing is implemented using a router on a stick configuration, while five vland isolate the office, shop, PYO kiosk, greenhouse IoT devices, and security cameras. Wired links support fixed devices such as PC's and printers, while access points provide wireless connectivity. Greenhouse IoT devices are placed in a dedicated vlan to prevent monitoring traffic affecting other services.

3.2 Network Topology

Figure 1 shows the topology of the proposed farm network. The design uses a hierarchical structure with one central router, one core switch, and four distribution switches serving the main farm areas.

Router0 performs inter VLAN routing through sub interfaces and connects to the core switch using a trunk link that carries multiple vlan. The core switch aggregates the access switches for the office, shop, PYO kiosk, and greenhouse networks, where each access switch connects local devices such as PC,s, printers, sensors, and access points. This structure improves scalability and simplifies management.

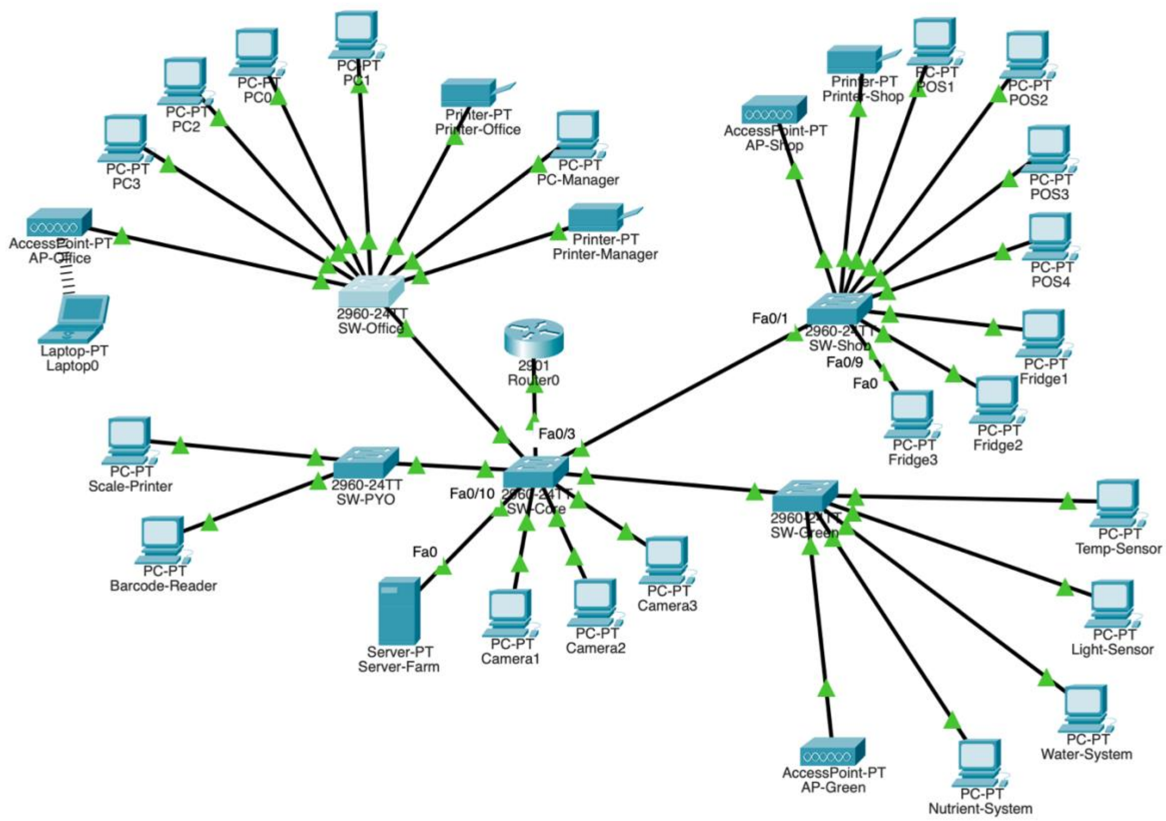

Figure 1: Full network topology of the farm network showing all vlans and connected devices

3.3 IP Addressing Scheme

A structured IP addressing scheme was used to separate the different farm network areas. Each vlan was assigned a private IPv4 subnet to simplify management and maintain consistent device configuration.
The default gateway for each vlan is configured on the router sub interfaces to support inter vlan communication. End devices use addresses from their assigned subnet and communicate through the corresponding gateway.

Table 1: IP addressing scheme for the farm network

Device	VLAN	IP Address	Subnet Mask	Default Gateway
Router0 GE0/0.10	10	192.168.10.1	255.255.255.0	
Router0 GE0/0.20	20	192.168.20.1	255.255.255.0	
Router0 GE0/0.30	30	192.168.30.1	255.255.255.0	
Router0 GE0/0.40	40	192.168.40.1	255.255.255.0	
Router0 GE0/0.50	50	192.168.50.1	255.255.255.0	
PC0	10	192.168.10.2	255.255.255.0	192.168.10.1
PC1	10	192.168.10.3	255.255.255.0	192.168.10.1
PC2	10	192.168.10.4	255.255.255.0	192.168.10.1
PC3	10	192.168.10.5	255.255.255.0	192.168.10.1
PC-Manager	10	192.168.10.6	255.255.255.0	192.168.10.1
Printer-Office	10	192.168.10.7	255.255.255.0	192.168.10.1
Printer-Manager	10	192.168.10.8	255.255.255.0	192.168.10.1
Laptop0 (wireless)	10	192.168.10.10	255.255.255.0	192.168.10.1
Server-Farm	10	192.168.10.100	255.255.255.0	192.168.10.1
POS1	20	192.168.20.2	255.255.255.0	192.168.20.1
POS2	20	192.168.20.3	255.255.255.0	192.168.20.1
POS3	20	192.168.20.4	255.255.255.0	192.168.20.1
POS4	20	192.168.20.5	255.255.255.0	192.168.20.1
Printer-Shop	20	192.168.20.6	255.255.255.0	192.168.20.1
Fridge1	20	192.168.20.7	255.255.255.0	192.168.20.1
Fridge2	20	192.168.20.8	255.255.255.0	192.168.20.1
Fridge3	20	192.168.20.9	255.255.255.0	192.168.20.1
Scale-Printer	30	192.168.30.2	255.255.255.0	192.168.30.1
Barcode-Reader	30	192.168.30.3	255.255.255.0	192.168.30.1

Temp-Sensor	40	192.168.40.2	255.255.255.0	192.168.40.1
Light-Sensor	40	192.168.40.3	255.255.255.0	192.168.40.1
Water-System	40	192.168.40.4	255.255.255.0	192.168.40.1
Nutrient-System	40	192.168.40.5	255.255.255.0	192.168.40.1
Camera1	50	192.168.50.2	255.255.255.0	192.168.50.1
Camera2	50	192.168.50.3	255.255.255.0	192.168.50.1
Camera3	50	192.168.50.4	255.255.255.0	192.168.50.1

3.4 VLAN Design

The network was logically segmented using Virtual Local Area Networks to improve security, reduce broadcast traffic, and simplify management. Each functional area of the farm was assigned to a dedicated vlan based on its role and traffic requirements.

vlan 10 supports the office network, vlan 20 is used for farm shop systems, vlan 30 serves the PYO kiosk, vlan 40 is dedicated to greenhouse IoT devices, and vlan 50 is reserved for security cameras. This separation ensures that traffic from one area does not unnecessarily affect other parts of the network.

Table 2: Vlan allocation in the farm network

VLAN ID	Name	Purpose	Subnet
10	Office	Staff PCs, printers, server	192.168.10.0/24
20	Shop	POS systems and fridges	192.168.20.0/24
30	PYO	Kiosk equipment	192.168.30.0/24
40	Greenhouse	IoT sensors and controllers	192.168.40.0/24
50	Security	CCTV cameras	192.168.50.0/24

```
Switch>enable
Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/9, Fa0/11, Fa0/12, Fa0/13
                                                Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                                Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                                Fa0/22, Fa0/23, Fa0/24, Gig0/2
10   Office                           active    Fa0/8, Fa0/10
20   Shop                             active
30   PYO                              active
40   Greenhouse                       active
50   Security                         active    Fa0/5, Fa0/6, Fa0/7
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#
```

Figure 2: VLAN configuration on the core switch showing all configured vlans

3.5 Wired Infrastructure

The wired infrastructure follows a hierarchical switched design. A core switch connects the central router to four distribution switches, each serving a specific farm area. Fixed devices like PC's, printers, cameras, and IoT controllers are connected using copper ethernet cables to provide stable and reliable connectivity.

Trunk links are configured between the core and distribution switches to allow multiple vlan to traverse the network while remaining logically separated at the access layer.

```
Switch#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      1
Fa0/2       on           802.1q         trunking      1
Fa0/3       on           802.1q         trunking      1
Fa0/4       on           802.1q         trunking      1
Gig0/1      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/1       1-1005
Fa0/2       1-1005
Fa0/3       1-1005
Fa0/4       1-1005
Gig0/1      1-1005

Port        Vlans allowed and active in management domain
Fa0/1       1,10,20,30,40,50
Fa0/2       1,10,20,30,40,50
Fa0/3       1,10,20,30,40,50
Fa0/4       1,10,20,30,40,50
Gig0/1      1,10,20,30,40,50

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       1,10,20,30,40,50
Fa0/2       1,10,20,30,40,50
Fa0/3       1,10,20,30,40,50
Fa0/4       1,10,20,30,40,50
Gig0/1      1,10,20,30,40,50
```

Figure 3: Trunk configuration on the core switch showing vlan traffic between switches

3.6 Wireless Infrastructure

Wireless connectivity was implemented to provide flexible access across the farm. Access points were deployed in the office, shop, and greenhouse to support mobile devices.

Each access point is connected to the appropriate switch and mapped to the correct vlan. This makes sure wireless clients receive IP addresses within their assigned network. For example, the office wireless network operates on vlan 10, allowing secure access to office resources.

Figure 4: Wireless laptop successfully connected to the office access point

3.7 IoT Integration

The greenhouse environment includes several IoT devices such as temperature, light, water, and nutrient monitoring systems. These devices are connected through SW-Green and isolated in vlan 40 to separate IoT traffic from user and business networks.
Each device is assigned a static IP address within the 192.168.40.0/24 subnet and uses 192.168.40.1 as the default gateway. As shown in Figure 5, the temperature

sensor communicates successfully with zero packet loss, confirming that the IoT segment is correctly configured.

.

```
C:\>ping 192.168.40.1

Pinging 192.168.40.1 with 32 bytes of data:

Reply from 192.168.40.1: bytes=32 time<1ms TTL=255
Reply from 192.168.40.1: bytes=32 time<1ms TTL=255
Reply from 192.168.40.1: bytes=32 time<1ms TTL=255
Reply from 192.168.40.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.40.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Figure 5: Successful connectivity test from greenhouse temperature sensor to its gateway (vlan 40)

3.8 Security Cameras

Three IP cameras were deployed and connected through the core switch using vlan 50. This isolates surveillance traffic from other operational areas such as the office, shop, and IoT systems.

Using a dedicated vlan improves both security and performance. If a camera is compromised, the threat remains contained within vlan 50. In addition, isolating high bandwidth video traffic helps prevent unnecessary load on other network segments.

3.9 Assumptions Made

Several assumptions were adopted to simplify the farm network simulation while keeping the design realistic. Physical cabling was assumed to support the required speeds, and all networking devices such as the router, switches, and access points were considered to operate normally. The number of connected devices in each area was treated as representative of typical farm needs, with room for future expansion within each vlan. The wireless environment was also assumed to have minimal interference, although real deployments may require further consideration of cable quality, hardware failures, and signal obstacles.

4 Analysis

4.1 Testing Results

After configuration, connectivity tests were performed in Packet Tracer as discussed before to verify network operation. These included gateway reachability, intra vlan communication, inter vlan routing, and server access.

PC0 successfully reached its default gateway at 192.168.10.1, confirming correct vlan 10 configuration. Successful replies were also received from the vlan 20, vlan 40, and vlan 50 gateways, verifying inter vlan routing. The farm server at 192.168.10.100 was reachable.

Test	Source Device	Destination	Result
Basic gateway connectivity	PC0	192.168.10.1	Success
Intra vlan communication	PC0	192.168.10.2	Success
Inter vlan to Shop	PC0	192.168.20.1	Success
Inter vlan to Greenhouse	PC0	192.168.40.1	Success
Inter vlan to Security	PC0	192.168.50.1	Success
Server connectivity	PC0	192.168.10.100	Success

```
C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Figure 6: Successful connectivity test from office PC to default gateway

```
C:\>ping 192.168.20.1

Pinging 192.168.20.1 with 32 bytes of data:

Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.20.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.40.1

Pinging 192.168.40.1 with 32 bytes of data:

Reply from 192.168.40.1: bytes=32 time=2ms TTL=255
Reply from 192.168.40.1: bytes=32 time<1ms TTL=255
Reply from 192.168.40.1: bytes=32 time<1ms TTL=255
Reply from 192.168.40.1: bytes=32 time=1ms TTL=255

Ping statistics for 192.168.40.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 2ms, Average = 0ms

C:\>ping 192.168.50.1

Pinging 192.168.50.1 with 32 bytes of data:

Reply from 192.168.50.1: bytes=32 time<1ms TTL=255
Reply from 192.168.50.1: bytes=32 time<1ms TTL=255
Reply from 192.168.50.1: bytes=32 time<1ms TTL=255
Reply from 192.168.50.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.50.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Figure 7: Inter VLAN routing test showing successful communication between network segments

```
C:\>ping 192.168.10.100

Pinging 192.168.10.100 with 32 bytes of data:

Reply from 192.168.10.100: bytes=32 time<1ms TTL=128
Reply from 192.168.10.100: bytes=32 time<1ms TTL=128
Reply from 192.168.10.100: bytes=32 time<1ms TTL=128
Reply from 192.168.10.100: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Figure 8: Successful connectivity test between office PC and farm server

```
C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.10.2: bytes=32 time=26ms TTL=128
Reply from 192.168.10.2: bytes=32 time=9ms TTL=128
Reply from 192.168.10.2: bytes=32 time=5ms TTL=128
Reply from 192.168.10.2: bytes=32 time=11ms TTL=128

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 5ms, Maximum = 26ms, Average = 12ms
```

Figure 9: Wireless laptop successfully communicating with the wired network

4.2 Design Evaluation

The implemented network design met the main requirements of the smart farm environment. vlan segmentation improved traffic organisation by separating key areas such as the office, shop, IoT greenhouse devices, and security cameras. This reduced unnecessary broadcast traffic and improved overall network structure.

Testing confirmed that inter vlan routing operates correctly when communication between segments is required. The hierarchical design using a core switch and distribution switches also supports scalability and simplifies management.

However, in a real deployment, the design could be enhanced by adding stronger security controls such as access control lists and improved wireless protection. Redundancy mechanisms could also be introduced to increase network availability.

4.3 Commercial Implications

From a commercial perspective, the proposed design balances cost and functionality. vlan segmentation allows multiple services to share the same infrastructure, reducing hardware needs and overall deployment cost.

Operational efficiency is also improved. Isolating greenhouse IoT devices supports better environmental monitoring, while the scalable structure allows future expansion with minimal redesign, improving return on investment.

4.4 Limitations

Despite the successful implementation, some limitations exist. The simulation was developed using Packet Tracer, which cannot fully represent real network conditions such as hardware failures, traffic load, or wireless interference.

The design also assumes stable connectivity and does not include advanced security controls. In real deployments, wireless coverage and IoT performance would require further optimisation.

5 Conclusion

This report presented the design and simulation of a smart farm network using Packet Tracer. The solution segmented the farm using vlans and enabled controlled communication through inter vlan routing. Both wired and wireless connectivity were implemented to support office users, retail systems, IoT devices, and security cameras.

Testing confirmed reliable communication within and between vlan’s. Overall, the design provides a scalable foundation for a modern smart farm. Future work may include stronger security controls, improved wireless coverage, and validation.

6 References

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Shaik, A., 2024. A survey of emerging techniques for large networks of virtual local area networks (vlans) with benefits and limitations. *International Journal of Current Engineering and Technology*, *14*(6), pp.478-487.

7 Appendices

Appendix A: Packet tracer file Link

https://drive.google.com/file/d/11Px5ok7tz1yba_d58IhY4Kz5HRDWvYbT/view?usp=share_link

Appendix B: Panopto link

https://southwales.cloud.panopto.eu/Panopto/Pages/Viewer.aspx?id=1a578524-ee32-4458-9126-b4010024b878

Appendix C

Key CLI Configuration

Router Configuration
interface GigabitEthernet0/0 no shutdown interface GigabitEthernet0/0.10 encapsulation dot1Q 10 ip address 192.168.10.1 255.255.255.0 interface GigabitEthernet0/0.20 encapsulation dot1Q 20 ip address 192.168.20.1 255.255.255.0 interface GigabitEthernet0/0.30 encapsulation dot1Q 30 ip address 192.168.30.1 255.255.255.0 interface GigabitEthernet0/0.40 encapsulation dot1Q 40 ip address 192.168.40.1 255.255.255.0 interface GigabitEthernet0/0.50 encapsulation dot1Q 50 ip address 192.168.50.1 255.255.255.0

Core Switch VLAN Configuration
vlan 10 name Office vlan 20 name Shop vlan 30 name PYO vlan 40 name Greenhouse vlan 50 name Security

Trunk Configuration (SW-Core)
interface GigabitEthernet0/1 switchport mode trunk interface range FastEthernet0/1-4 switchport mode trunk

Access Port Examples
interface range FastEthernet0/2-9 switchport mode access switchport access vlan 10 spanning-tree portfast

Appendix D

Simulation Evidence

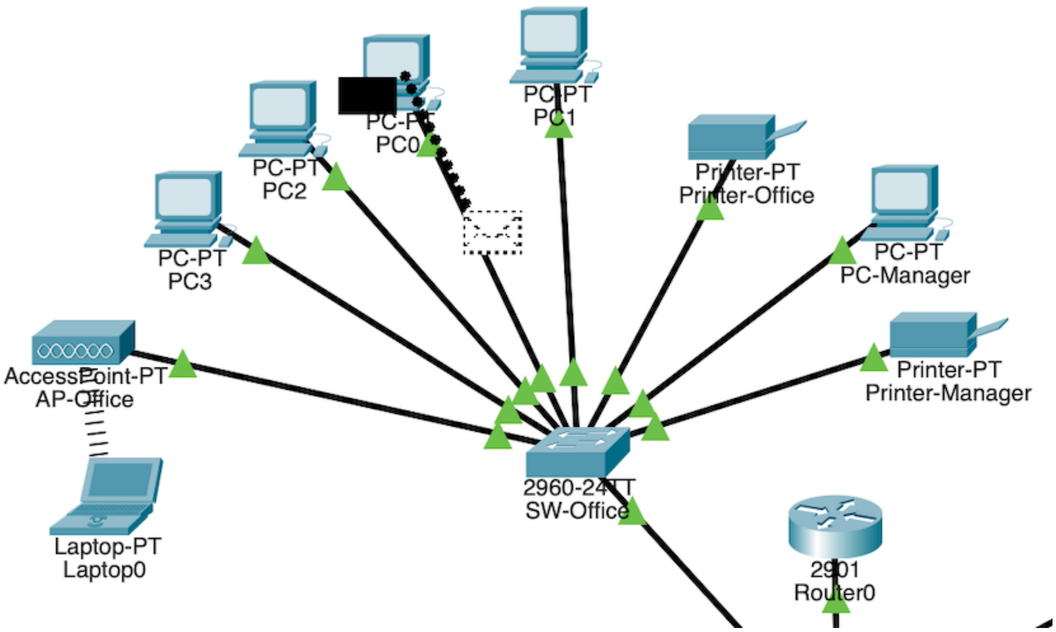

Figure 10: Simulation Evidence